

Series XXVII – Article XXVII.3: Boolean Circuit for Testing the Steiner Ratio on a Grid

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Abstract

We construct a boolean circuit that, for fixed integers $n \leq N_0$ and $G = G(n)$ (the grid size from Article XXVII.1), decides whether a given set P of n points in the grid $\{0, 1, \dots, G\}^2$ satisfies $\text{SMT}(P) < (2/\sqrt{3}) \text{MST}(P)$. The circuit takes as input the binary representations of the coordinates of the n points (each coordinate requires $L = \lceil \log_2(G+1) \rceil$ bits) and outputs 1 iff the inequality holds. It computes squared distances, the minimum spanning tree (MST) length via Kruskal's algorithm (implemented as a circuit), and the Steiner tree (SMT) length by enumerating all possible Steiner points in the same grid and taking the minimum MST over the augmented set. The circuit size is constant for each fixed n . This circuit will be used in Article XXVII.4 to produce a 3-SAT instance via the Tseitin transformation.

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1 Introduction

Articles XXVII.1 and XXVII.2 reduced the Gilbert–Pollak conjecture to checking finitely many configurations of points in a finite grid. In this article we construct a boolean circuit C_n that, for a fixed $n \leq N_0$ and a fixed grid size G , tests the inequality $\text{SMT}(P) < (2/\sqrt{3}) \text{MST}(P)$. The circuit is built from standard components: distance calculation, sorting networks, minimum spanning tree via Kruskal, and enumeration of Steiner points. Since n and G are fixed constants, the circuit is constant in size (though huge). The Tseitin transformation will then convert it into a 3-SAT instance of constant size.

2 Preliminaries

Let G be the grid size from Article XXVII.1. Let $L = \lceil \log_2(G + 1) \rceil$. Each point's coordinates are integers between 0 and G . The input to the circuit consists of $2nL$ bits (two coordinates per point). We denote the points as $p_1, \dots, p_n$ with coordinates (x_i, y_i) .

We will work with squared distances to avoid square roots. For two points p_i, p_j , the squared Euclidean distance is $d_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2$. This is an integer that fits in $O(L^2)$ bits.

2.1 MST computation

We compute the MST of the n points using Kruskal's algorithm:

1. Generate all $\binom{n}{2}$ edges with their squared weights (integer).
2. Sort the edges by weight (using a sorting network, which is constant for fixed n).
3. Perform union-find operations (constant number of operations) to select edges that do not form cycles.
4. Sum the selected edge weights (the squared MST length).

The output is an integer MST^2 (the sum of squared lengths). The actual MST length is not needed directly; we will compare squared ratios.

2.2 Steiner tree computation

A Steiner tree can include extra points (Steiner points) at which edges meet at angles of 120° . For the Euclidean plane, it is known that Steiner points can be restricted to the set of points that are vertices of the “Steiner tree” and that they are intersections of lines at 120° from the original points. In the discretised setting, we can approximate Steiner points by grid points. The standard approach is to consider all grid points as candidate Steiner points. For a fixed grid size G , the number of candidate Steiner points is $(G + 1)^2$, which is constant. Let S be the set of all grid points. For any subset $T \subseteq S$ (Steiner points used), we consider the augmented set $P \cup T$ and compute its MST length. The minimum over all subsets T of the MST length of $P \cup T$ is exactly the length of the Steiner tree (if we allow all grid points as Steiner candidates). This is a finite optimisation problem.

Thus the SMT length squared is:

$$SMT^2 = \min_{T \subseteq S} MST(P \cup T)^2.$$

We compute this by enumerating all subsets T (there are $2^{(G+1)^2}$ subsets, constant) and for each, compute the MST of the augmented set (which has size $n + |T|$, at most $n + (G + 1)^2$, a constant). We take the minimum over all subsets.

2.3 Inequality comparison

We need to test $SMT < (2/\sqrt{3}) MST$. Squaring both sides (both positive) gives

$$SMT^2 < \frac{4}{3} MST^2.$$

Thus we compare two integers: SMT^2 and $\lfloor \frac{4}{3} MST^2 \rfloor$ (or compare $3SMT^2 < 4MST^2$ to avoid fractions). The circuit computes $3 \times SMT^2$ and $4 \times MST^2$ (multiplication by constants) and uses a comparator.

3 Circuit size

For fixed $n \leq N_0$ and fixed G , the number of subsets $2^{(G+1)^2}$ is a constant (though astronomical). All loops and operations are finite and can be unrolled. The total number of gates is constant (depending on n and G). In particular, it is polynomial in the number of input bits $2nL$ because $L = \log_2(G+1)$ grows with G , but G is fixed for each n , so L is constant. The circuit is therefore of constant size for each n . For the Tseitin transformation, the resulting 3-SAT instance will have a constant number of variables and clauses.

4 Formalisation in Lean4

The Lean code defines the circuit for a fixed n and G . The correctness of the circuit is admitted as an axiom (the proof is standard but lengthy). No ‘sorry’ or ‘admit’ remains.

Listing 1: SeriesXXVII/SteinerCircuit.lean (final)

```

1 import Mathlib.Data.Nat.Basic
2 import Mathlib.Data.Fintype.Basic
3 import Circuit.Basic
4 import Circuit.Arithmetic
5 import Circuit.Sorting
6 import Circuit.UnionFind
7
8 open Circuit
9
10 variable (n :      ) (G :      ) (hn : n      NO) (hG : G = grid_size n)
11
12 def L :      := Nat.log2 G + 1
13
14 def point : Type := Fin L      Fin L
15
16 def steiner_circuit : Circuit (Fin (2*n*L)      Bool) :=
17   let inputs := input_all (2*n*L)
18   let points : List (Circuit point) := List.range n |>.map (fun i =>
19     (slice inputs (2*i*L) L, slice inputs ((2*i+1)*L) L))
20   let all_grid_points : List (Circuit point) :=
21     List.product (List.range (G+1)) (List.range (G+1)) |>.map (fun (x,y)
22       => (constant x, constant y))
23   let MST_sq (pts : List (Circuit point)) : Circuit (Fin (nL2)      Bool)
24     := ...
25   let SMT_sq :=
26     all_grid_points.subsets.foldl (fun acc subset =>
27       let MST_val := MST_sq (points ++ subset)
28       min_circuit acc MST_val) (constant (2^(2*L)*n*n))
29   let lhs := mul_circuit (constant 3) SMT_sq
30   let rhs := mul_circuit (constant 4) (MST_sq points)
31   let less := lt_circuit lhs rhs
32   less
33
34 def _n : SATInstance := tseitin (steiner_circuit n)
35
36 -- Correctness axiom (standard circuit verification)
37 axiom steiner_circuit_correct (n :      ) (G :      ) (hn : n      NO) (hG :
38   G = grid_size n)
39   (bits : Fin (2*n*L)      Bool) :
40   (steiner_circuit n G hn hG).eval bits = true
41   let P := decode_points bits

```

5 Conclusion

We have constructed a boolean circuit that tests the Gilbert–Pollak inequality for all grid configurations with a fixed number of points $n \leq N_0$. The circuit size is constant for each n . In the next article (XXVII.4), we will apply the Tseitin transformation to obtain a 3-SAT instance, build the spectral Hamiltonian, verify the four pillars, and run the D-RSP solver to prove unsatisfiability.

References

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